

Question 1. (1.5 points)

An automobile manufacturer introduces a new model that averages 27 miles per gallon in the city. A person who plans to purchase one of these new cars wrote the manufacturer for the details of the tests, and found out that the standard deviation is 3 miles per gallon. Assume that in-city mileage is approximately normally distributed.

- a) What is the probability that the person will purchase a car that averages less than 20 miles per gallon for in-city driving?
- b) What is the probability that the person will purchase a car that averages between 25 and 29 miles per gallon for in-city driving?
- c) If 10 cars tested. Find the probability that at least 7 cars that averages between 25 and 29 miles per gallon for in-city driving.

Question 2. (1.5 points)

A company self-insures its large fleet of cars against collisions. To determine its mean repair cost per collision, it has randomly chosen a sample of 16 accidents. If the average repair cost in these accidents is \$2,200 with a sample standard deviation of \$800, find a 90 percent confidence interval estimate of the mean cost per collision.

Question 3. (2.0 points)

Of 100 randomly detected cases of individuals having lung cancer, 67 died within 5 years of detection.

- a) Estimate confidence interval for proportion that a person contracting lung cancer will die within 5 years at 90%.
- b) How large an additional sample would be required to be 95 percent confident that the error in estimating the probability in part a) is less than .02?

Question 4. (1.5 points)

The number of hours worked per week by medical residents is normally distributed with a standard deviation of 4.8 hours. We want to test the claim that the mean number of hours worked per week by medical residents is more than 60 hours. Based on a sample of 36 medical residents, what is the largest value of the sample mean for which we would fail to reject the null hypothesis at a 5% significance level?

Question 5. (2.0 points)

A manufacturer of interocular lenses will qualify a new grinding machine if there is evidence that the percentage of polished lenses that contain surface defects does not exceed 2%. A random sample of 250 lenses contains 6 defective lenses.

- a) Formulate and test an appropriate set of hypotheses to determine whether the machine can be qualified. Use $\alpha = 0.05$.
- b) Find the P-value.

Question 6. (1.5 points)

The viscosity of two different brands of car oil is measured and the following data resulted:

Brand 1: 10.62, 10.58, 10.33, 10.72, 10.44, 10.74

Brand 2: 10.50, 10.52, 10.58, 10.62, 10.55, 10.51, 10.53

Is there enough statistical evidence to show that the mean viscosity of the two brands are equal assuming that the populations have normal distributions with equal variances with significance level 5%.